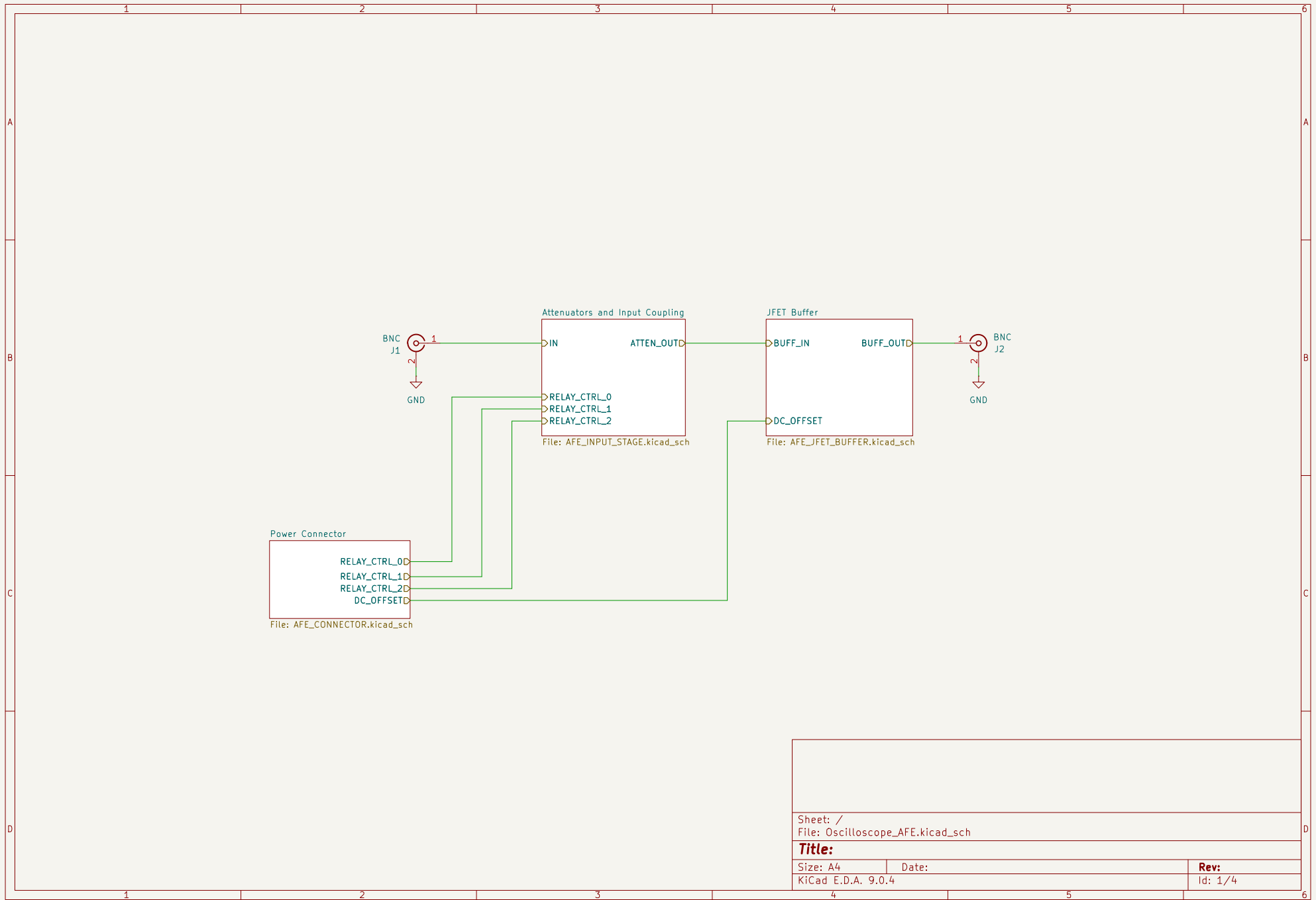




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INPUT COUPLING AND ATTENUATORS

The schematic diagram illustrates the input coupling and attenuators for the AFE_INPUT_STAGE. It features three input channels (IND, RELAY_CTRL_0D, RELAY_CTRL_1D, RELAY_CTRL_2D) connected to a common input stage. The input stage includes AC/DC coupling, impedance path selection, and attenuation. The output is connected to a common output stage.

AC/DC Coupling: The input signal (IND) is connected to the AC_PATH through a 0.1uF capacitor (C1). The DC_PATH is connected to the input signal through a 0.1uF capacitor (C1).

Impedance Path Select: The input signal is connected to the 50Ω Path or 1MΩ Path through a switch (K1A, K1B, K2A, K2B).

Attenuators: The input signal is connected to the 100:1 Attenuator or 10:1 Attenuator through a switch (K3A, K3B).

Decoupling Capacitors: The input signal is connected to the +5V and -5V supply rails through decoupling capacitors (C6, C7, C8).

Relay Control: The input signal is connected to the RELAY_CTRL_0D, RELAY_CTRL_1D, and RELAY_CTRL_2D through relays (Q1, Q2, Q3).

Output: The output signal is connected to the DATTEN_OUT through a switch (K2B).

Components:

- Resistors: R1 (990k), R2 (10k), R3 (900k), R4 (100k), R5 (75), R6 (120), R7 (75), R8 (5), R9 (5), R10 (5).
- Capacitors: C1 (0.1u), C2 (10p), C3 (10p), C4 (990p), C5 (90p), C6 (0.1u), C7 (0.1u), C8 (0.1u).
- Diodes: D1 (1N914WT), D2 (1N914WT), D3 (1N914WT), D4 (BAT54T1G), D5 (BAT54T1G).
- Relays: K1A, K1B, K2A, K2B, K3A, K3B, K1C (G6J-2FL-Y DC5), K2C (G6J-2FL-Y DC5), K3C (G6J-2FL-Y DC5).
- Transistors: Q1 (2N7002), Q2 (2N7002), Q3 (2N7002).

Test Points: TP1, TP2.

Decoupling Capacitors: C6 (0.1u), C7 (0.1u), C8 (0.1u).

Relay Control: RELAY_CTRL_0D, RELAY_CTRL_1D, RELAY_CTRL_2D.

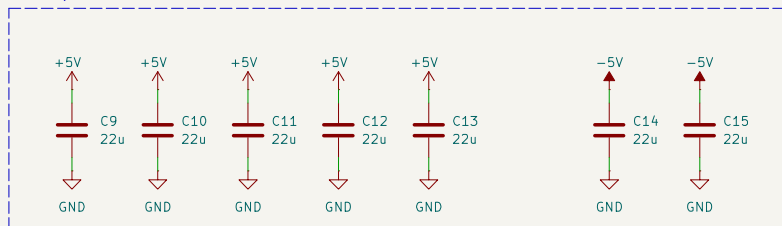
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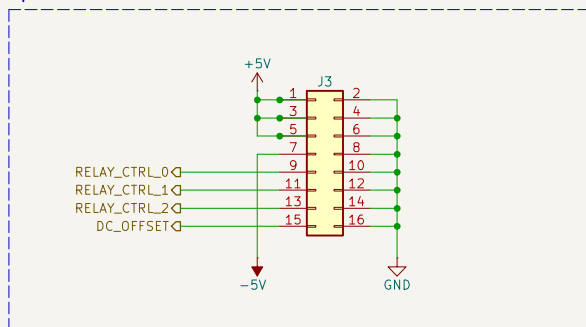
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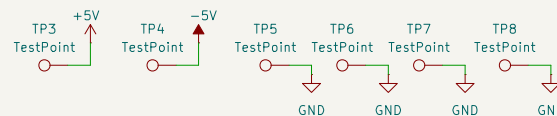
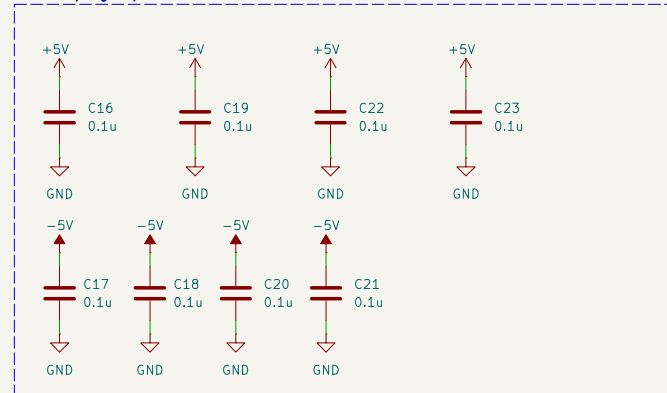
Bulk Capacitors



Input Connector



IC Decoupling Capacitors



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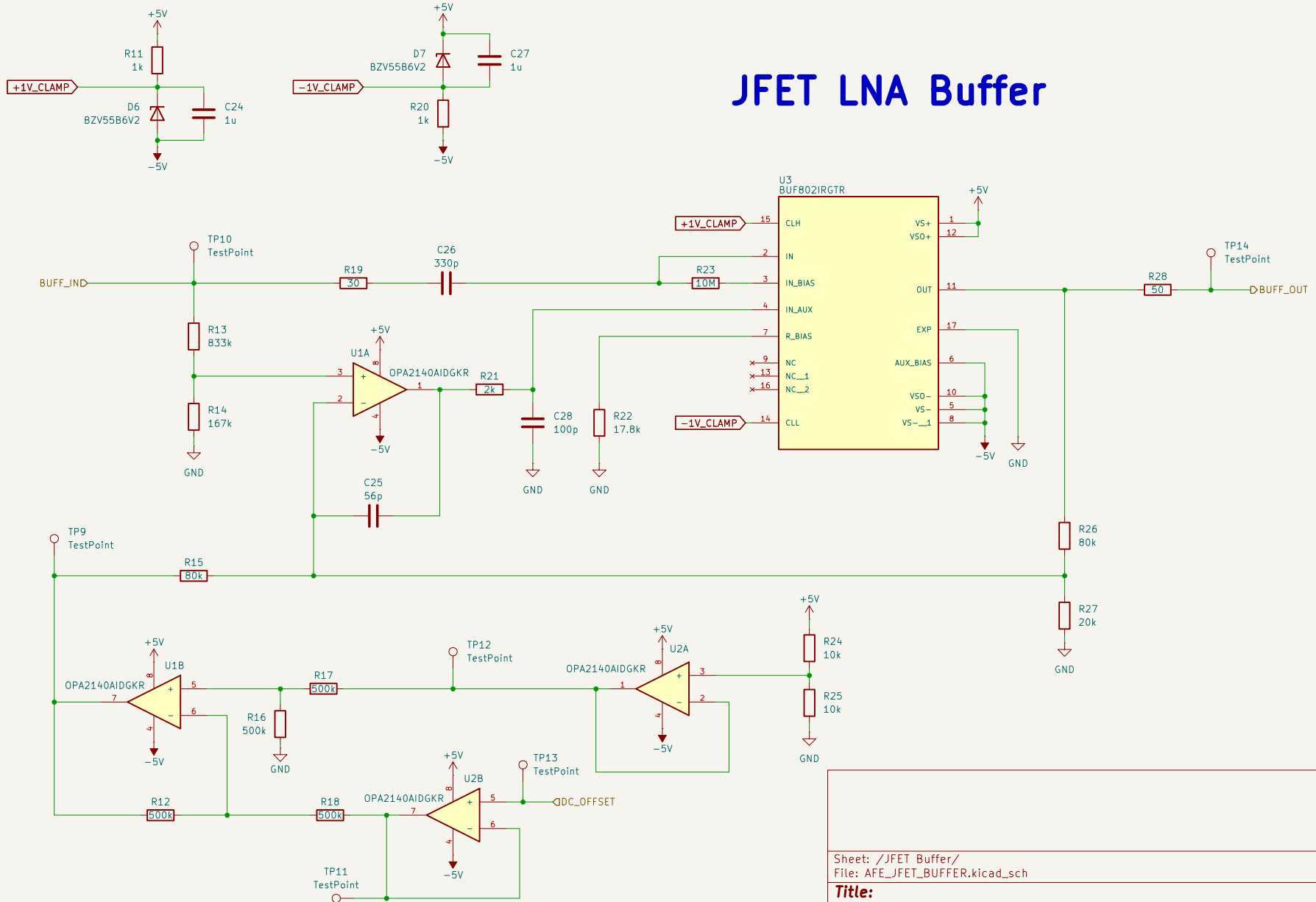
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JFET LNA Buffer



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